Star Digital: Advertising Effectiveness Analysis

Presented By Group 1

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Introduction and Business Context

Online display advertising is one of the main channels through which Star Digital reaches potential customers and drives online conversions. However, the firm currently faces uncertainty about the true causal effectiveness of its advertising and whether the observed conversions can be attributed to advertising exposure rather than underlying customer differences.

In this project, we analyze data from an experiment in which some users are assigned to see online ads (treatment group) while others do not see the ads (control group). This experimental setting allows us to compare purchase behavior across the two groups and to assess the causal impact of online advertising on conversions.

Our objective is to provide Star Digital with evidence-based guidance on whether its online advertising is effective and how it should allocate its advertising budget across different publisher sites. Specifically, we address four questions:
(1) Is online advertising effective in increasing conversion rates compared to the control group?
(2) Does advertising frequency, measured by the number of impressions, have a positive and potentially nonlinear effect on the probability of purchase?
(3) How does the conversion effectiveness of Sites 1–5 compare with that of Site 6?
(4) Given the relative performance of these sites, how should Star Digital allocate its future advertising budget across Sites 1–5 and Site 6?

Data Description and Experimental Design

To answer these questions, we use descriptive statistics, hypothesis testing, and causal modeling. We first compare raw conversion rates between the treatment and control groups and conduct a difference-in-means test. We then estimate a series of logistic regression models to quantify the effect of advertising exposure, the frequency of impressions, and site-specific impressions on purchase probabilities. Throughout the analysis, we interpret results through a causal inference lens and discuss the assumptions under which these estimates can be viewed as causal effects of online advertising.

Data Overview

The dataset contains 57,681 observations tracking individual user sessions, with key variables including purchase (binary outcome), test (treatment/control indicator), and impressions across six publisher sites (imp_1 through imp_6). Treatment users were randomly assigned to see ads, while control users were shown charity ads instead of Star Digital ads. Summary statistics show an overall conversion rate of approximately 50%, with treatment impressions varying substantially across users and sites. This high conversion rate is due to choice-based sampling, where buyers are oversampled for analysis. In reality, the true conversion rate is only 0.153%. Since users were randomly assigned to

treatment and control groups, the difference in purchase rates can be interpreted as the causal effect of advertising.

Q1. Effectiveness of Online Advertising

To evaluate whether online advertising is effective, we first compare conversion rates between the treatment and control groups. As shown in Figure 1, in the control group (no ad exposure), the conversion rate is 48.6%, whereas in the treatment group (exposed to the ad) the conversion rate is 50.5%. This implies a raw difference of about 1.9 percentage points in favor of the treatment group.

We formally test this difference using a two-sample t-test of the mean purchase rates. The test yields a p-value of 0.061 and a 95% confidence interval for the difference in means of approximately [-0.039, 0.001]. Therefore, at the conventional 5% significance level we cannot reject the null hypothesis of equal conversion rates, although there is weak evidence of a positive effect at the 10% level.

We also estimate a simple logistic regression of purchase on the treatment indicator. The estimated coefficient on the treatment variable is 0.077, corresponding to an odds ratio of about 1.08. This suggests that treated users have roughly 8% higher odds of purchasing than control users, holding all else constant in this simple specification.

Taken together, these results indicate that online advertising appears to have a small and only marginally significant positive effect on purchase behavior, so the evidence for a strong advertising effect is suggestive rather than conclusive.

These estimates can be interpreted as causal effects of ad exposure if assignment to the treatment group is effectively random and uncorrelated with unobserved user characteristics. Because users were randomly assigned to treatment and control groups, this difference can be interpreted as the causal effect of advertising, not just correlation.

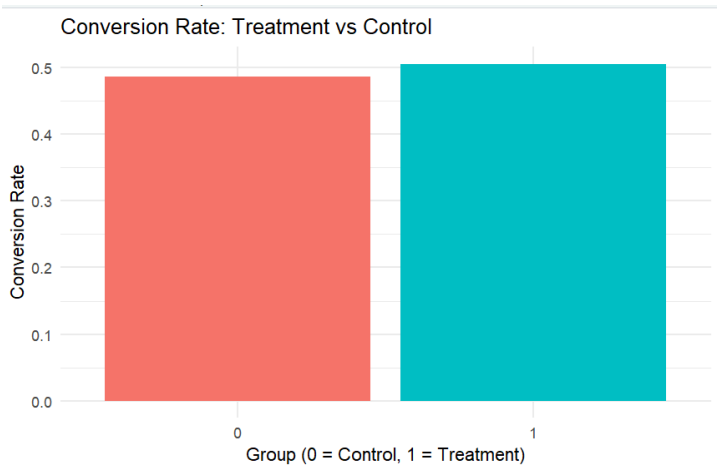

Figure 1: Conversion Rate Comparison (Treatment vs Control)

Q2. Frequency Effect of Advertising Impressions
To examine whether advertising frequency affects purchase behavior, we first estimate a logistic regression of purchase on the total number of ad impressions across all six sites.

The coefficient on total impressions is 0.029 ($p < 0.001$), corresponding to an odds ratio of about 1.03. This implies that each additional impression is associated with roughly a 3% increase in the odds of purchase, indicating a clear positive frequency effect.

We then allow for a nonlinear relationship by including a squared term for total impressions. In this quadratic specification, the linear term remains positive and highly significant, while the squared term is negative and highly significant. As shown in Figure 2, purchase probability initially increases with additional impressions, but the marginal benefit diminishes at higher levels of exposure and may eventually decline.

Figure 2 plots the predicted purchase probability against total impressions based on the quadratic model. The curve rises steeply at low to moderate levels of exposure, flattens out as impressions accumulate, and turns downward at very high impression counts.

Overall, these results provide strong evidence of a frequency effect: more impressions generally increase the likelihood of purchase, but there are diminishing returns and potential over-exposure at the upper end of the impression distribution. From a managerial perspective, this means that showing too many ads can reduce efficiency, so firms should cap the number of impressions per user.

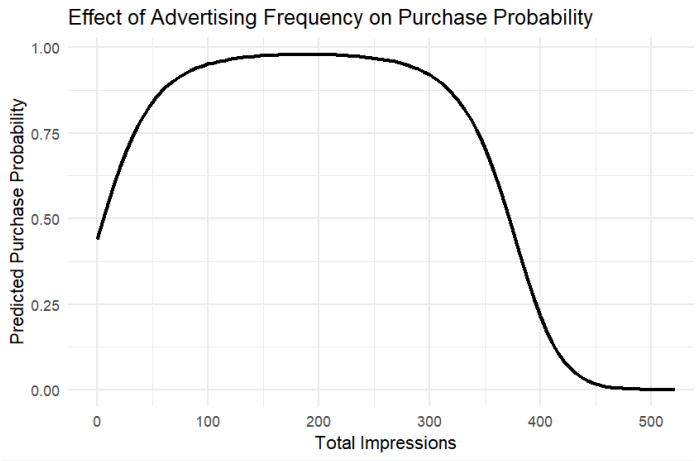

Figure 2: Predicted Purchase Probability vs Total Impressions

Q3. Site-Level Effectiveness (Sites 1–6)
To compare the effectiveness of the different publisher sites, we estimate a logistic regression of purchase on the number of impressions received on each site (imp_1 through imp_6).

The resulting odds ratios show substantial heterogeneity across sites. Impressions on Site 4 are the most effective, with an odds ratio of about 1.20, indicating that each additional impression on Site 4 increases the odds of purchase by roughly 20%. Impressions on Site 2 and Site 6 are also positively

associated with conversion, with odds ratios slightly above 1, although their effects are more modest in magnitude.

In contrast, several sites appear to perform poorly. Impressions on Site 3 and especially Site 5 have odds ratios below 1 and statistically significant negative coefficients, suggesting that additional impressions on these sites are associated with lower odds of purchase. Impressions on Site 1 have an odds ratio very close to 1 and are not statistically significant, implying little to no effect on conversion.

Overall, as shown in Figure 3, these results suggest that Site 4 is clearly the strongest performer, Sites 2 and 6 provide moderate incremental value, while Sites 3 and 5 are underperforming and may not be good candidates for additional investment. These results should be interpreted with caution, as site-level estimates may still be influenced by unobserved user characteristics (e.g., higher-intent users visiting certain sites), and therefore may not fully represent causal effects.

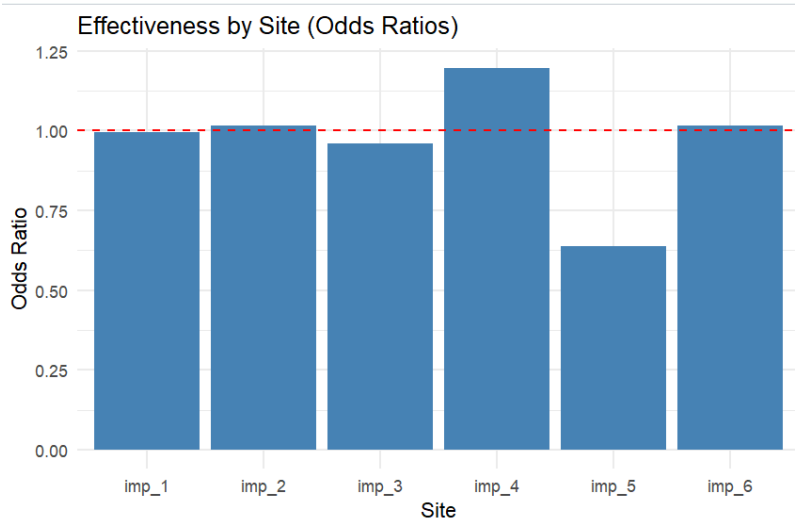

Figure 3: Odds Ratios by Publisher Site

Q4. Budget Allocation Recommendations

To determine how Star Digital should allocate its advertising budget, we estimate the incremental conversions generated by Sites 1–5 and Site 6, and use these to compute return on investment (ROI). This approach isolates the incremental impact of advertising by comparing predicted outcomes with and without ad exposure.

We decompose ad exposure into two components: an initial effect (whether a user saw at least one ad) and a frequency effect (the impact of additional impressions). We then estimate a logistic regression model including the treatment indicator, exposure variables (show_1_5, freq_1_5, show_6, freq_6), and interaction terms between treatment and exposure. These interaction terms capture the incremental lift caused by Star Digital's advertising.

The regression coefficients used to estimate incremental lift are shown in Figure 4.

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Coefficients:
  (Intercept)           test       show_1_5       freq_1_5         show_6         freq_6
    -0.389282      -0.016514       0.447295       0.014537      -0.087997       0.007807
test:show_1_5  test:freq_1_5    test:show_6    test:freq_6
     0.038027       0.010797       0.019876       0.025195
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Figure 4: Logistic Regression Coefficients with Interaction Terms

Because the dataset uses choice-based sampling (50% purchases compared to the true rate of 0.153%), we apply an offset adjustment to align predicted probabilities with real-world purchase rates. This ensures that our results are representative of actual customer behavior.

Next, we generate counterfactual predictions to estimate incremental conversions attributable to each site. These conversions are multiplied by the average purchase value ($1,200) to estimate incremental revenue. Finally, using advertising costs ($25 per 1,000 impressions for Sites 1–5 and $20 per 1,000 impressions for Site 6), we compute ROI for each option.

The ROI results are summarized in Figure 5

Option <chr>	Impressions <dbl>	Cost <dbl>	Incremental_Conversions <dbl>	Incremental_Revenue <dbl>	ROI <dbl>
Sites 1-5	138034	3450.85	33.65956	40391.47	10.70479
Site 6	40177	803.54	10.38501	12462.01	14.50889

Figure 5: ROI Comparison for Sites 1–5 vs Site 6

As shown in Figure 5, Site 6 delivers a higher return on investment than Sites 1–5. This is primarily due to its stronger frequency effect and lower cost per impression.

Interestingly, Sites 1–5 generate a larger initial impact, meaning they are more effective at first exposure. In contrast, Site 6 has a stronger frequency effect, indicating that repeated impressions on Site 6 are more effective at driving conversions.

This suggests a complementary strategy: Sites 1–5 are better for building initial awareness, while Site 6 is more effective for reinforcing the message through repeated exposure.

Therefore, we recommend a hybrid budget allocation strategy, where Star Digital uses Sites 1–5 to generate initial awareness and allocates more follow-up impressions to Site 6 to maximize overall return on investment.

These results incorporate both advertising effectiveness and cost, providing a more complete and practical basis for budget allocation decisions.

Key Takeaways and Managerial Implications

Our analysis provides several important insights for Star Digital's advertising strategy.

First, online display advertising increases purchase probability, although the effect is modest and only marginally significant. This suggests that advertising is effective, but its impact should be interpreted with realistic expectations.

Second, we find strong evidence of a frequency effect: additional impressions increase the likelihood of purchase, but with diminishing returns at higher levels of exposure. This implies that there is an optimal range of impressions, beyond which additional advertising becomes less efficient.

Third, advertising effectiveness varies significantly across publisher sites. Site 4 stands out as the strongest performer in terms of per-impression effectiveness, while Sites 3 and 5 appear to be underperforming. Site 6 shows a positive impact and plays an important role in driving conversions through repeated exposure.

When both effectiveness and cost are considered, Site 6 delivers the highest return on investment due to its lower cost per impression and stronger frequency effect.

Based on these findings, we recommend a hybrid advertising strategy. Star Digital should use high-performing sites such as Site 4 to generate initial awareness, while allocating more follow-up impressions to Site 6 to reinforce messaging and maximize overall return on investment.

Additionally, the company should carefully manage advertising frequency to avoid over-exposure and ensure that impressions remain within the range of positive marginal returns.

By combining effective site selection with strategic frequency management, Star Digital can improve the efficiency and overall performance of its advertising investments.

These recommendations are based on causal estimates from a randomized experiment, providing a reliable foundation for decision-making.

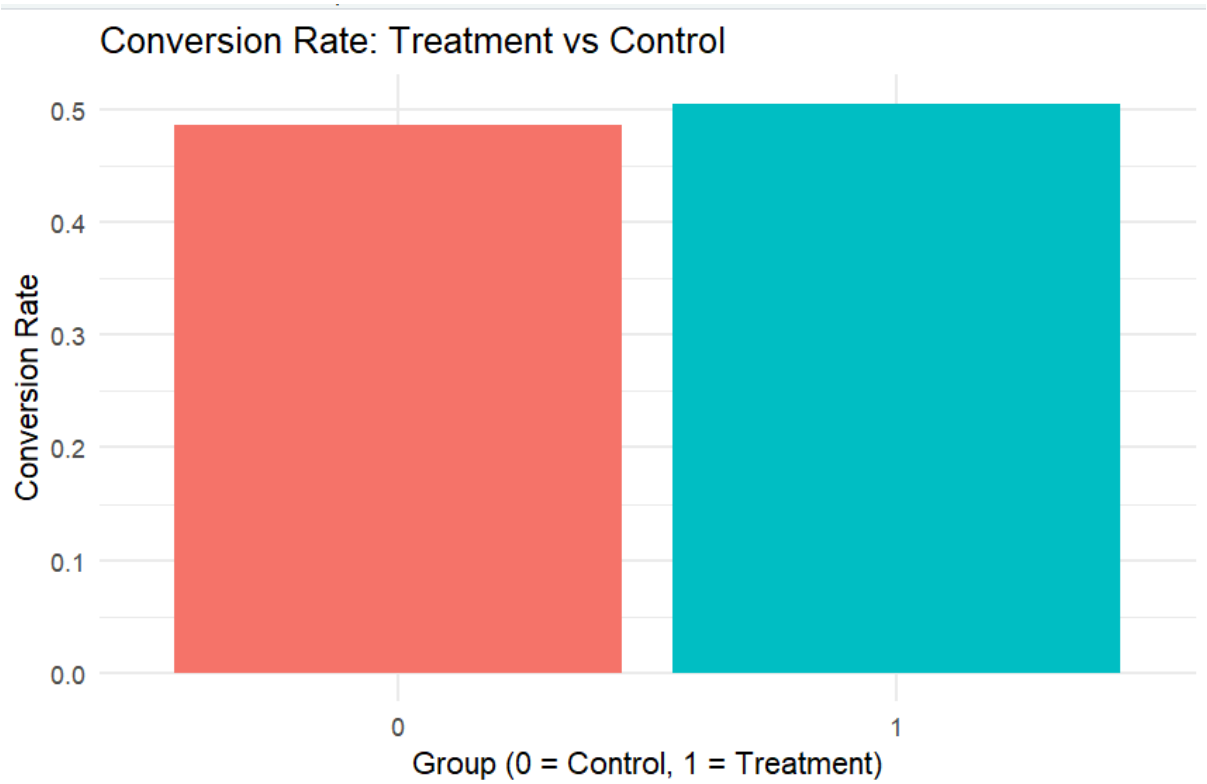
Overall, this analysis demonstrates how experimental data and causal modeling can be combined to make data-driven marketing decisions, allowing firms to move beyond correlation-based metrics and toward true measurement of advertising effectiveness.

References

- Dataset: Star Digital advertising experiment (57,681 user sessions)
- Statistical software: R version 4.3.2
- Models: Logistic regression with binomial family

Appendix: Figures

Figure 1: Conversion Rate Comparison (Treatment vs Control)



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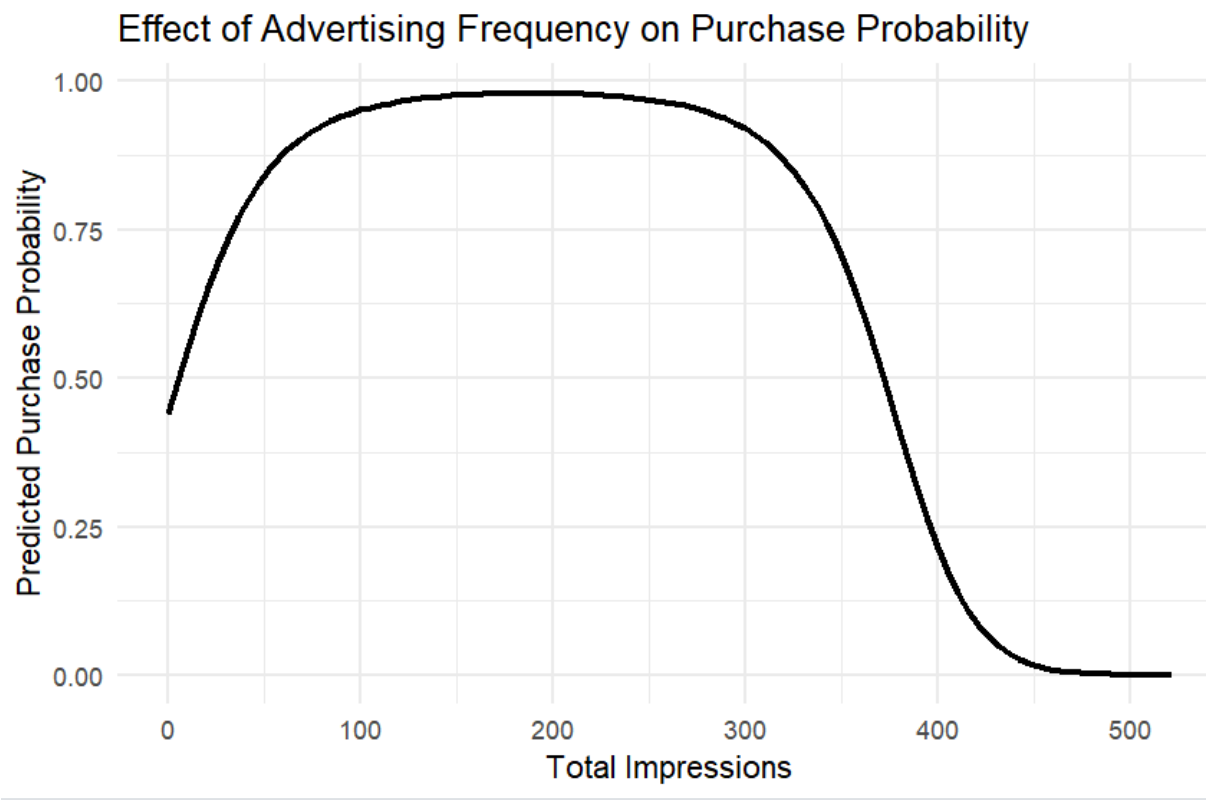

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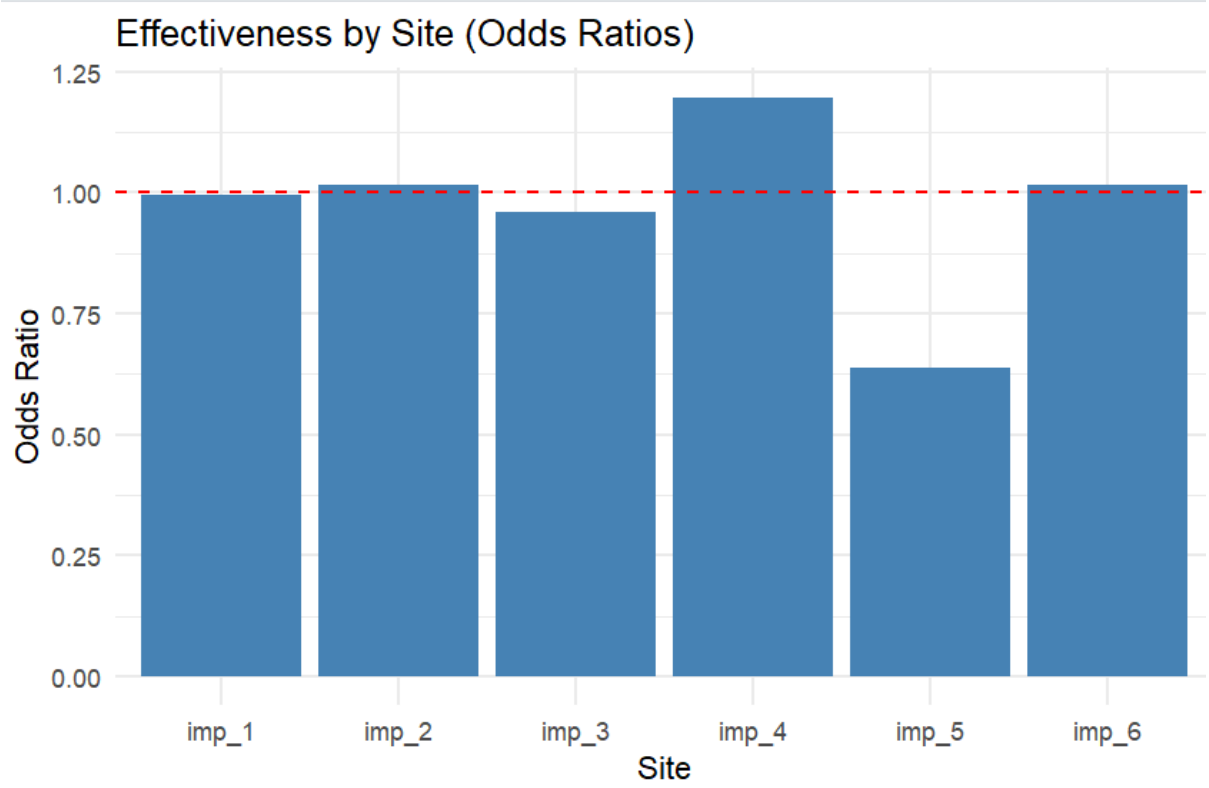
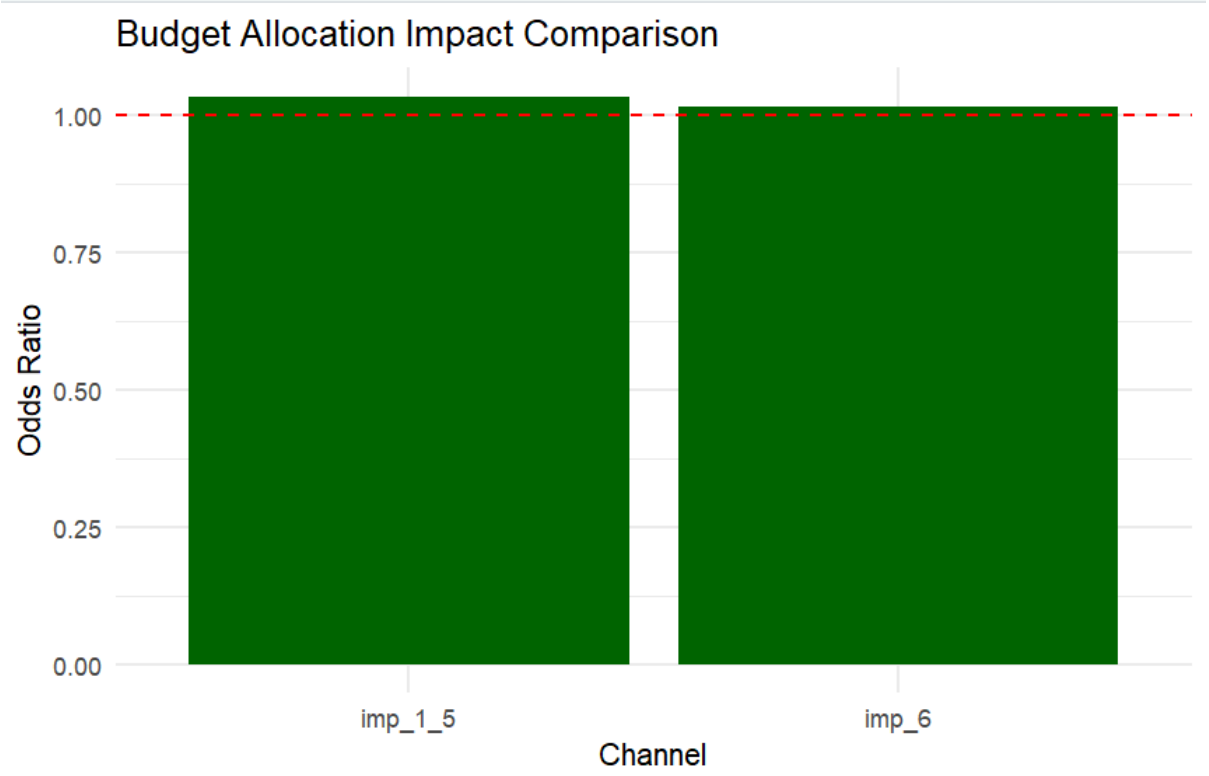
Figure 4 Budget Allocation Comparison